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AI CAREERS FOR WOMEN

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Pay-Lens: visualizing salary truth with AI & BI

Project Domain

[Excel / Power BI]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

Student Name
Sneha khandelwal

Student Email
Snehakhandelwal666@gmail.com

Guide
R. Umamaheshwari

Institution: [H.L College of commerce]

Year - 2026

3. Problem Statement

Background of the Chosen Domain and Data-Related Challenges

The project focuses on Human Resources (HR) and Payroll Management, a domain where organizations handle vast amounts of sensitive compensation data. Currently, this data is often raw, inconsistent, or locked in complex payroll systems. A primary challenge is the Information Asymmetry between the employer and the employee; while companies use advanced tracking, employees—especially gig and blue-collar workers—lack clear, verifiable access to how their wages are calculated.

Significance in a Real-World Context

This problem is critical because it involves Legal Compliance and Financial Stability. Under Indian labour laws (such as the Factories Act), overtime must be paid at 2x the ordinary rate. In the real world, poor transparency often leads to systemic underpayment of overtime or incorrect statutory deductions (PF, Professional Tax) that go unnoticed by the workforce. Furthermore, pay inequity—where similar roles have widely different packages—erodes organizational trust and fairness.

Specific Gap or Inefficiency Addressed

The project aims to bridge the **"Transparency and Skills Gap"**:

- **The Transparency Gap:** There is a lack of clear tools to help employees understand the conversion from Gross CTC to In-hand salary.
- **The Skills Gap:** Most HR professionals lack the Prompt Engineering skills needed to use AI tools like Microsoft Copilot to automate the detection of payroll discrepancies.
- **The Analytical Gap:** There is often no automated "Ratio Analysis & interpretation" to compare internal salaries against Market Median benchmarks.

Who is Affected and the Value Delivered

- **Affected Parties:** Blue-collar workers and gig employees face the highest risk of wage loss. HR departments are affected by high turnover costs, which typically equal 1.5x the annual salary of the departing employee.
- **Value Delivered:** The solution provides Automated Verification to ensure legal compliance, reducing the risk of lawsuits and penalties. It delivers Financial Literacy to employees through clear deduction breakdowns, improving morale and retention

4. Objectives

- To collect and explore the raw payroll dataset (1,000 records) to understand salary structures and data quality.

- To clean and preprocess data by handling missing values and standardizing formats for CTC, Fixed, and Variable pay.

- To perform Ratio Analysis by comparing internal salaries with Market Median benchmarks (25th, 50th, 75th percentiles).

- To build an Interactive Power BI Dashboard that visualizes the relationship between salary increases, tenure, and legal compliance.

- To integrate AI tools for generating automated discrepancy alerts when overtime pay falls below legal standards. [free AI tools can be applied in this part]

5. Methodology / Implementation Plan

sr .no	Phase	Implementation Details
1	**Data Collection & Exploration**	Obtain the 1,000-record payroll dataset from Kaggle including Employee Master, Attendance, and Market Benchmark tables. We will identify key fields such as CTC_Annual, Market_Median_Salary, and Overtime Hours to understand the baseline for transparency.
2	**Data Cleaning & Preprocessing**	Use Excel Power Query to handle null values in Employee_ID, fix date formats for Date_of_Joining, and standardize text. For the AI component, we will draft structured prompts using the Role-Context-Task-Format-Tone framework to prepare for discrepancy analysis.
3	**Analysis**	Implement Excel formulas and DAX measures in Power BI to calculate the Market Positioning Index and verify 2x Overtime pay compliance. For financial literacy part free AI tools can be integrated
4	**Visualization / Output**	Develop an interactive Power BI dashboard featuring KPI cards for "Total Payroll Cost" and "Salary vs. Market Ratio." The output will include slicers to filter by department (e.g., IT, Sales) to identify specific areas of pay inequity.
5	**Documentation & Submission**	Compile the final project report, document all AI prompts used, and upload the complete repository to GitHub. A video demonstration will be recorded to showcase the dashboard's ability to flag legal non-compliance.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i5-12400/AMD Ryzen 5 5500U or higher, minimum DDR5 8 GB RAM (DDR5 16 GB recommended)
- Operating System: Windows 11 or macOS 13 Ventura
- Microsoft Excel (Microsoft 365 / Excel 2019 or higher)
- Power BI Desktop — latest version (free download from Microsoft)
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

1) 1,000-record CSV containing employee IDs, Department, CTC, and Market Benchmarks. (multiple datasets sourced from Kaggle and rest of the column created by excel financial formulas)
2) https://www.kaggle.com/datasets?search=employee
3) After collecting different data set from Kaggle following steps need to done:
   a) Define the columns required
   b) Apple Excel formulas like
      i) To create a unique employee ID: ="EMP"&TEXT(ROW(A1),"000")
      ii) To allot the dept: create a department list in worksheet -- use a formula given below by selecting the dept list reference
      =INDEX(Z1: Z6,RANDBETWEEN(1,6))
      iii) To allot random date of joining:
      =RANDBETWEEN(DATE(2019,1,1),DATE(2025,12,31))
      iv) To calculate CTC and all the financial calculation can use following formulas
      =E2*12% [employee PF]
      =E2*40% [HRA]
      =E2*30% [allowances]
      =SUM(E2:H2) [total CTC]
      =I2*12 [annual CTC]
      Note: these all formulas are on the basis of my dataset and I have created the dataset using this formula.

C. Project Timeline

A table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*

2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

- **Intelligent Payroll Data Infrastructure:**
  Development of a clean, standardized, and audit-ready payroll dataset (1,000+ records), enabling reliable financial analysis and decision-making within HR systems.

- **Automated Legal Alerts:** A system that successfully flags any underpayment of **Overtime (2x rate)**, ensuring compliance with Indian Labor Laws (Factories Act).

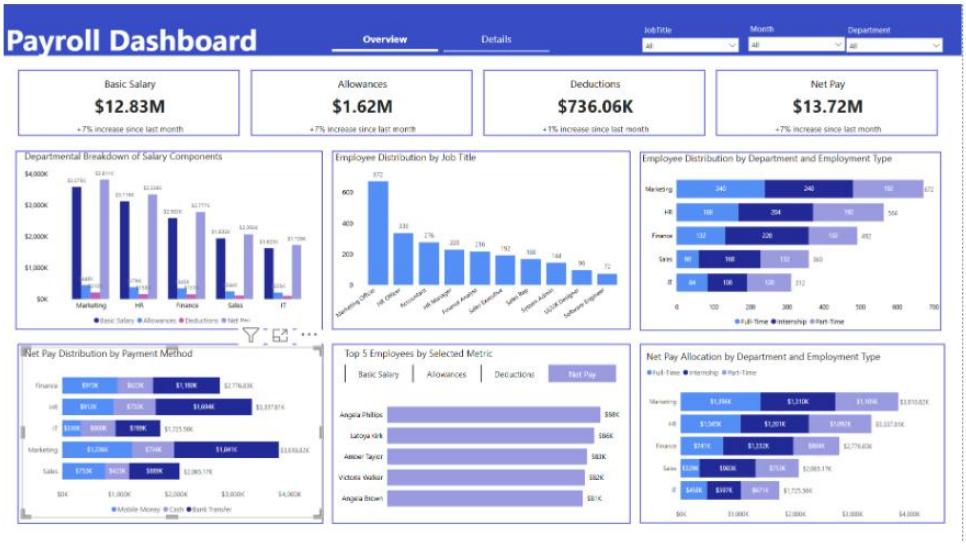
- **Quantitative Pay Equity Framework:**
  Creation of a structured **Market Positioning Index** to evaluate internal salary distributions against industry benchmarks, enabling organizations to detect pay disparities and optimize compensation strategies.

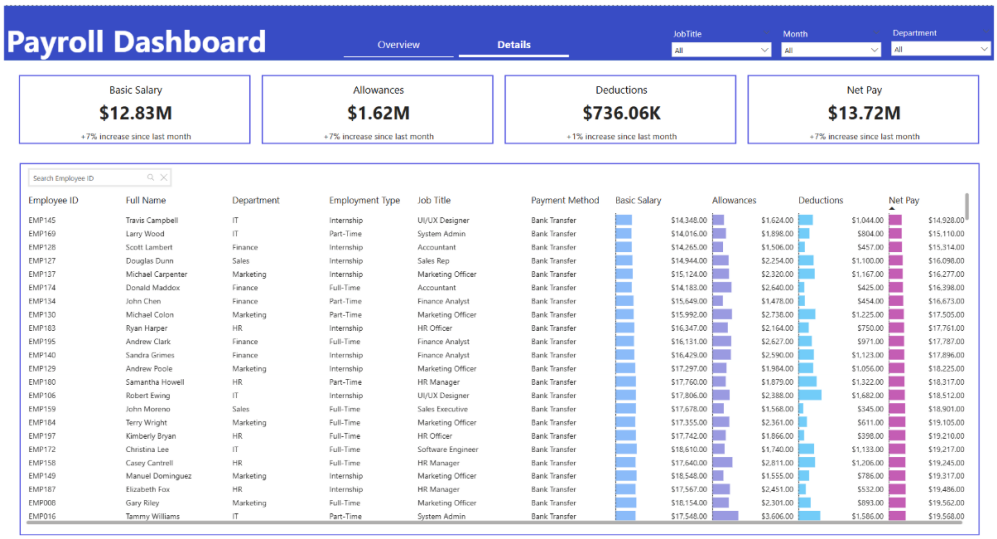
- **Enhanced Financial Transparency for Workforce:**
  Transformation of complex salary structures into **simplified, AI-assisted explanations**, empowering employees—especially gig and blue-collar workers—with clarity on deductions, benefits, and net pay.

- **Decision-Intelligence Dashboard:**
  Development of an interactive Power BI dashboard that converts raw payroll data into **actionable insights**, highlighting trends in compensation, tenure, departmental cost structures, and potential attrition risks.

- **AI-Augmented HR Analytics Capability:**
  Integration of prompt-based AI tools to assist HR professionals in detecting anomalies, interpreting financial ratios, and automating repetitive analytical tasks, thereby reducing manual effort and improving accuracy.

- **Bridging the Transparency–Analytics Gap:**
  The project establishes a scalable framework that combines **data analytics, AI, and business intelligence** to address systemic inefficiencies in payroll systems and improve organizational trust.





Employee ID	Full Name	Department	Employment Type	Job Title	Payment Method	Basic Salary	Allowances	Deductions	Net Pay
EMP145	Travis Campbell	IT	Internship	UI/UX Designer	Bank Transfer	$14,348.00	$1,624.00	$1,044.00	$14,928.00
EMP169	Larry Wood	IT	Part-Time	System Admin	Bank Transfer	$14,016.00	$1,898.00	$804.00	$15,110.00
EMP128	Scott Lambert	Finance	Internship	Accountant	Bank Transfer	$14,265.00	$1,506.00	$457.00	$15,314.00
EMP127	Douglas Dunn	Sales	Internship	Sales Rep	Bank Transfer	$14,944.00	$2,254.00	$1,100.00	$16,098.00
EMP137	Michael Carpenter	Marketing	Internship	Marketing Officer	Bank Transfer	$15,124.00	$2,320.00	$1,167.00	$16,277.00
EMP174	Donald Maddox	Finance	Full-Time	Accountant	Bank Transfer	$14,183.00	$2,640.00	$425.00	$16,398.00
EMP134	John Chen	Finance	Part-Time	Finance Analyst	Bank Transfer	$15,649.00	$1,478.00	$454.00	$16,673.00
EMP130	Michael Colon	Marketing	Part-Time	Marketing Officer	Bank Transfer	$15,992.00	$2,738.00	$1,225.00	$17,505.00
EMP163	Ryan Harper	HR	Internship	HR Officer	Bank Transfer	$16,347.00	$2,164.00	$750.00	$17,761.00
EMP195	Andrew Clark	Finance	Full-Time	Finance Analyst	Bank Transfer	$16,131.00	$2,627.00	$971.00	$17,787.00
EMP140	Sandra Grimes	Finance	Internship	Finance Analyst	Bank Transfer	$16,429.00	$2,590.00	$1,123.00	$17,896.00
EMP129	Andrew Poole	Marketing	Internship	Marketing Officer	Bank Transfer	$17,297.00	$1,984.00	$1,056.00	$18,225.00
EMP180	Samantha Howell	HR	Part-Time	HR Manager	Bank Transfer	$17,760.00	$1,879.00	$1,322.00	$18,317.00
EMP106	Robert Ewing	IT	Internship	UI/UX Designer	Bank Transfer	$17,806.00	$2,388.00	$1,682.00	$18,512.00
EMP159	John Moreno	Sales	Full-Time	Sales Executive	Bank Transfer	$17,678.00	$1,568.00	$345.00	$18,901.00
EMP184	Terry Wright	Marketing	Full-Time	Marketing Officer	Bank Transfer	$17,355.00	$2,361.00	$611.00	$19,105.00
EMP197	Kimberly Bryan	HR	Full-Time	HR Officer	Bank Transfer	$17,742.00	$1,866.00	$398.00	$19,210.00
EMP172	Christina Lee	IT	Full-Time	Software Engineer	Bank Transfer	$18,610.00	$1,740.00	$1,133.00	$19,217.00
EMP158	Casey Cantrell	HR	Full-Time	HR Manager	Bank Transfer	$17,640.00	$2,811.00	$1,206.00	$19,245.00
EMP149	Manuel Dominguez	Marketing	Internship	Marketing Officer	Bank Transfer	$18,548.00	$1,555.00	$786.00	$19,317.00
EMP187	Elizabeth Fox	HR	Internship	HR Manager	Bank Transfer	$17,567.00	$2,451.00	$532.00	$19,486.00
EMP008	Gary Riley	Marketing	Full-Time	Marketing Officer	Bank Transfer	$18,154.00	$2,301.00	$893.00	$19,562.00
EMP016	Tammy Williams	IT	Part-Time	System Admin	Bank Transfer	$17,548.00	$3,606.00	$1,586.00	$19,568.00

Note: these are the visuals taken from the GitHub, link is mentioned in the references. After doing all the task on dataset, my dashboard will look like this above-mentioned dashboard.

8. References

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org
- Kaggle Inc. Open Datasets. Available at: https://www.kaggle.com/datasets?search=employee

- GitHub - evitanegara/Payroll-Dashboard-using-Power-BI: Interactive Power BI dashboard to analyze payroll distribution, employee salary trends, department-wise costs, and compensation insights.
- Indian Labour Laws (Factories Act)
- https://share.google/HryLSyGEVVxL70jfB : The Gig Workers' Strike Shows that Sustained Pressure from the Labor Force Does Yield Results - The Wire [an article]
- https://youtu.be/Xik34jh-doc?si=2y_bsf5HYzC3aDrA : YouTube link for the problem statement arise
- https://share.google/cNryzGFmCiB0W0gLH : [an article]
- Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel
- Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

Organizations across industries collect large volumes of data on a daily basis. However, this data is often raw, inconsistent, incomplete, or poorly structured, making it difficult to derive meaningful insights. Decision-makers rely on accurate, clean, and well-visualized data to make informed choices, but without the right tools and skills, this data remains underutilized.

Additionally, the rapid growth of Artificial Intelligence and Generative AI tools such as Microsoft Copilot and AI Chatbots has created both an opportunity and a skills gap. Professionals who can effectively use these tools through well-structured prompts can significantly accelerate their work. However, most users lack the knowledge of prompt engineering techniques needed to unlock the full potential of these AI tools.

This project addresses these challenges by applying data cleaning, analysis, visualization, and AI integration techniques using industry-standard tools. The project will demonstrate how structured data workflows combined with AI assistance can produce accurate, actionable insights and outputs relevant to a chosen real-world domain.

Students should include in their write-up:

- background of the chosen domain and the data-related challenges it faces
- Why this problem is significant in a real-world context
- What specific gap or inefficiency the project aims to address
- Who is affected and what value the solution will deliver

4. Objectives

List the specific goals your project aims to accomplish. Objectives should be clear and measurable.

- To collect and explore the raw dataset, understand its structure, fields, and data quality issues.
- To clean and preprocess the data by removing duplicates, handling missing values, correcting formats, and standardizing fields.
- To apply domain-relevant analysis using the assigned tool (Excel formulas and pivot tables, Power BI measures, or AI prompt techniques).
- To build an interactive dashboard, structured report, or AI-generated output that communicates insights clearly to stakeholders.
- To integrate AI tools (Microsoft Copilot or AI Chatbot) to automate, enhance, or accelerate the analysis and content generation process.
- To document the complete project process, upload all files to GitHub, and record a demonstration video for submission.

[Add any project-specific objective that is unique to your domain or chosen tool here.]

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description
1	**Data Collection & Exploration**	Obtain the raw dataset (CSV / Excel) from Kaggle, Edunet, or any suitable source. Understand field names, data types, and row count. For AI prompt projects, define the use case and target audience instead.
2	**Data Cleaning & Preprocessing**	Remove duplicates, handle null or missing values, fix incorrect data types, and standardize date and text formats using Excel (Remove Duplicates, IF, IFERROR, Text-to-Columns) or Power Query. For prompt-based projects, draft and structure initial prompts using Role, Context, Task, Format, and Tone components.
3	**Analysis & AI Integration**	Apply pivot tables, Excel formulas (SUMIF, COUNTIF, VLOOKUP), or Power BI DAX measures to derive key insights. Where applicable, use Microsoft Copilot or an AI Chatbot to generate narratives, summaries, or content outputs from the data or prompts.
4	**Visualization / Output**	Build interactive dashboards with charts, KPI cards, and slicers in Power BI or Excel. For AI-assisted projects, compile generated content outputs such

		as reports, marketing copy, or summaries. Document all AI prompts used alongside their outputs.
5	**Documentation & Submission**	Write the project report, upload all files to a GitHub repository, and record a video demonstration. Submit the GitHub link, video link, and dashboard link (if applicable) for evaluation.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Power BI Desktop — latest version (free download from Microsoft) — where applicable
- Microsoft Copilot — via Microsoft 365 subscription — where applicable
- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

- Whether the dataset is already available or needs to be sourced
- Source link if publicly available (e.g., Kaggle, UCI Repository, Edunet-provided)
- If not available, how the dataset will be collected, generated, or simulated
- For AI prompt projects: description of the use case brief and reference content used as input

7. Expected Outcomes / Conclusion

This section summarizes what the team expects to deliver by the end of the project. Outcomes should be realistic and directly tied to the objectives stated above.

- A clean, validated dataset with all preprocessing steps documented and a before/after comparison.

- An interactive dashboard or structured analytical report that clearly communicates key insights from the data.
- Demonstrated use of an AI tool (Microsoft Copilot or AI Chatbot) to generate insights, summaries, or content outputs, with prompts and results documented.
- A GitHub repository containing all project files, a completed fellowship project report, and a video demonstration of the working output.

[Add any outcome specific to your project domain here — for example: a prompt library, a published Power BI dashboard, a marketing content portfolio, or health metrics visualizations.]

8. References

List all references, tutorials, and documentation used in the project. Students must cite any material they referred to, including research papers, GitHub repositories, YouTube tutorials, blogs, and official documentation.

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: https://www.linkedin.com/learning
- Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel
- Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/
- Microsoft Corporation. Microsoft Copilot. Available at: https://copilot.microsoft.com
- Kaggle Inc. Open Datasets. Available at: https://www.kaggle.com/datasets
- *[Add your project-specific references here: research papers, GitHub links, YouTube tutorials, blogs, or any other material consulted.]*